Math 54 Discussion Worksheet

Name: ______________________________ ID: ____________________

Problem 1

Let A be any 4×4 matrix with eigenvalues $5, -3$, and 2. Suppose you know that the eigenspace associated to the eigenvalue $\lambda_1 = 5$ is 2-dimensional. Must A be diagonalizable?

Problem 2

a) The trace of a square matrix A is the sum of its diagonal elements and is denoted by $\mathrm{Tr}(A)$. It can be verified that $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ (and in fact trace is cyclic: $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB) = \mathrm{Tr}(BCA)$). Show that if A and B are similar, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$.

b) Prove that if A is diagonalizable, then the trace is just the sum of the eigenvalues of A.

Problem 3

Let A be a real symmetric matrix (all entries are real and $A^T = A$). Define $q = \overline{x}^T Ax$. Show that q is real (i.e. $\overline{q} = q$).Then, show that if $Ax = \lambda x$, then λ must be real and the real part of x is also an eigenvector of A.